

Business Brief: AI Call Center Optimization for CBRE

The Problem

CBRE runs call centers that handle building maintenance requests across their commercial real estate portfolio. Think water leaks, broken elevators, busted doors and windows. They already tried automating the process but the current system gets it wrong constantly. The biggest issue is that it sends low severity problems to 911. There is no human checkpoint to catch these mistakes before they go out, so bad calls just go through unchecked.

What We Know

From the kickoff we nailed down a few things. The root of the problem is the automated priority classification. It looks like the current setup is some kind of hybrid system where calls get transcribed and then auto classified, but that classification step is where everything breaks. We confirmed that transcripts are available to us as a data source. We also know CBRE has a ticketing system on the backend that our solution would need to feed into. And we agreed that it is reasonable to assume there are already existing codes or categories for the most common maintenance problems.

Proposed Direction

The solution should center on an NLP/LLM agent that reads call transcripts and pulls out the important pieces. What is the problem, where is it, and how urgent is it. From there it classifies the issue against the existing category codes. The key thing the current system is missing is a confidence threshold. When the model is not sure or when the situation looks serious it should flag the case for a human before anything gets dispatched. On the tech side a LangGraph agent with RAG over historical calls fits well here. It can pull similar past cases to make better classification decisions and use tool calls to push tickets into the existing system. The transcript data does double duty as both a retrieval index and an evaluation set to benchmark accuracy.

Open Questions

There are still a few gaps. We do not have a concrete accuracy target or success metric from the client yet. We also need a better sense of call volume and how many buildings or regions this would cover. The exact format of the transcript data has not been confirmed either which will affect how fast we can start prototyping. Lastly we know emergency scenarios exist but we have not fully mapped out which edge cases absolutely require a human versus which ones can be fully automated.